

Turning Price Swings into Savings:

Smart Load-Shifting Service in Finland

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1. Introduction

Since the 2022 energy crisis, around one in three Finnish power contracts refers to the hourly spot price (Korhonen, 2025). Fully electric automobiles rose by 39% in 2025 to 169,458 (Statistics Finland, 2026). Wind capacity was 9.4 GW, 36% of it in Northern Ostrobothnia near Oulu (Renewables Finland, 2026). Prices fluctuated dramatically within each day.

This study assesses one opportunity: a service automatically shifting household power use (EV charging, water heating) to the cheapest hours, starting from Oulu. It's a feasibility analysis, that sees the possibility viable only if four standards are met:

1. **Value:** Prices vary by a few cents per kWh in a normal day.
2. **Persistence:** the swing is being held every year, not just during the energy crisis.
3. **Predictability:** inexpensive hours are predictable and scheduled so software can act on them.
4. **Customer benefit:** average home saves around 15 € every month.

2. Panopto link to the video

Stakeholder presentation:

https://oulu.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=1731bb96-6494-435a-9c55-b4e90145121f

3. Opportunity and Data

3.1 Why this opportunity

I chose this scenario for three reasons. It’s local and timely: Oulu has lengthy winters, which boosts demand, and the area has a large share of Finland’s wind power. In addition, it can be assessed objectively with publicly available data before an investment. And the service is basically a scheduling mechanism, connecting Business Analytics with my computer science background.

3.2 Dataset and preparation

I utilized the hourly dataset from the open-source project nordpool-predict-fi (vividfog, 2026) which combines Nord Pool day-ahead prices (via Sähkötin), Fingrid wind and nuclear production, and Finnish Meteorological Institute observations, including Oulu Airport temperature. I choose it because it puts prices and their physical drivers from official sources in one table, at hourly resolution, with 32,853 records over 45 full months (January 2023 – September 2026). Monthly data would disguise the intra-day swings on which the opportunity rests.

Cleaning made the data correct and comparable: I removed 170 forecasts rows and the incomplete October 2026; confirmed that there were no duplicate or missing hours; converted UTC to Finnish time; removed VAT using the official rates (10%, 24%, 25.5%), because the rate changed twice; excluded ten hours at −500 €/MWh caused by a trader's bidding error on 24 November 2023 (Bloomberg, 2023); and set 44 negative wind readings to zero. Each step is recorded in the Excel file.

4. Analysis and Findings

4.1 Methods and visual choices

The methods are deliberately simple and transparent, suited to a go/no-go decision: descriptive statistics, pivot-style aggregation by day, month, hour and season, banding by wind and temperature, correlation, and a what-if household simulation. Python handled cleaning; all summaries are recalculating Excel formulas. Line charts show trends and daily rhythms, bar charts compare groups, and direct labels make each chart readable without prior knowledge of the data.

4.2 Price swings are large and persistent

The average price for the day was just 5.0 c/kWh, while the difference between the costliest and least expensive hour was 7.8 c/kWh (Figure 1). The energy crisis has not dampened the swing, it peaked in 2025 (Table 1). Moving 1 kWh from the average daily use to the four cheapest hours of the day saved around 2.8 cents per kWh shifted.

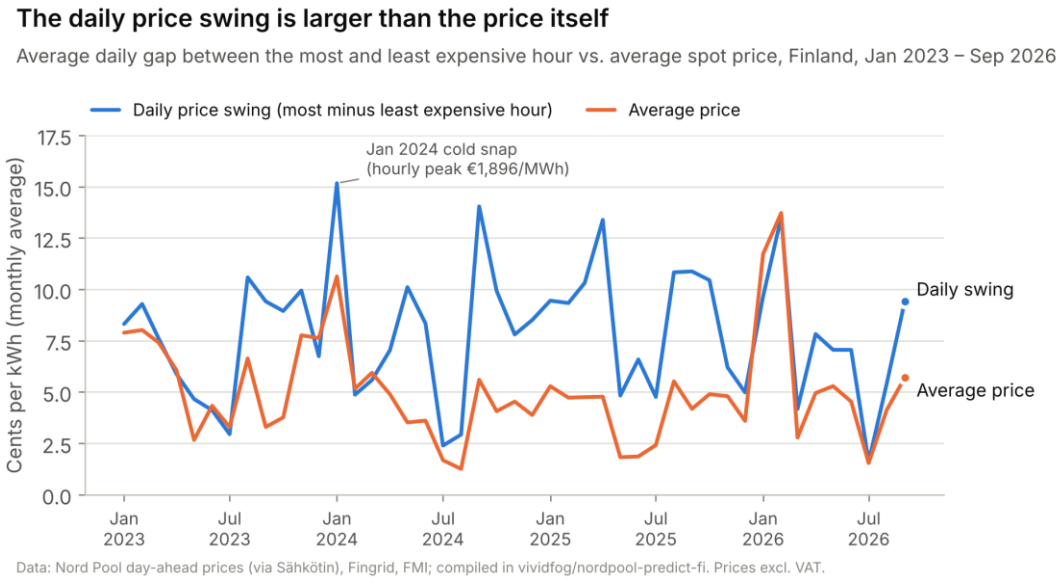


Figure 1. *Daily price swing versus average price, by month.*

Table 1. *Key metrics by period.*

Metric	2023	2024	2025	Last 12 months
Average price, c/kWh	5.71	4.56	4.05	5.58
Average daily swing, c/kWh	7.35	8.06	8.48	7.22
Negative-price hours	457	726	465	104
Saving per kWh shifted, c	2.80	2.74	2.85	2.81
Household saving, €/year (incl. VAT)	172	180	185	181

4.3 Cheap hours are predictable

Prices follow a consistent rhythm in every season: lowest between 00–06, high around 09 and 19 (Figure 2) Day-ahead prices are released the afternoon prior to allow for scheduling loads without forecasting. The oscillations are driven by wind and cold (Figure 3). Hours with less than 1 GW of wind had an average of 7.9 c/kWh, against 1.2 c/kWh above 5 GW (r = −0.38). Days when Oulu was below −20 °C had swings of 23.4 c/kWh, three times a usual day. The competition is also getting tougher for fixed night timers. The number of days where one of the four cheapest hours was outside 22-07 increased from 32% (2023) to 53% (2026).

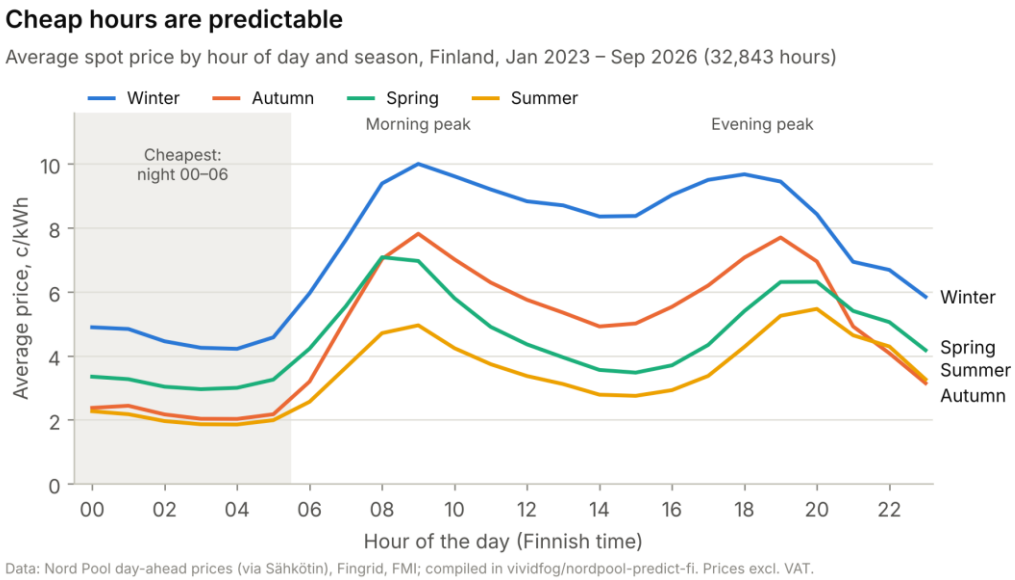


Figure 2. *Average price by hour of day and season.*

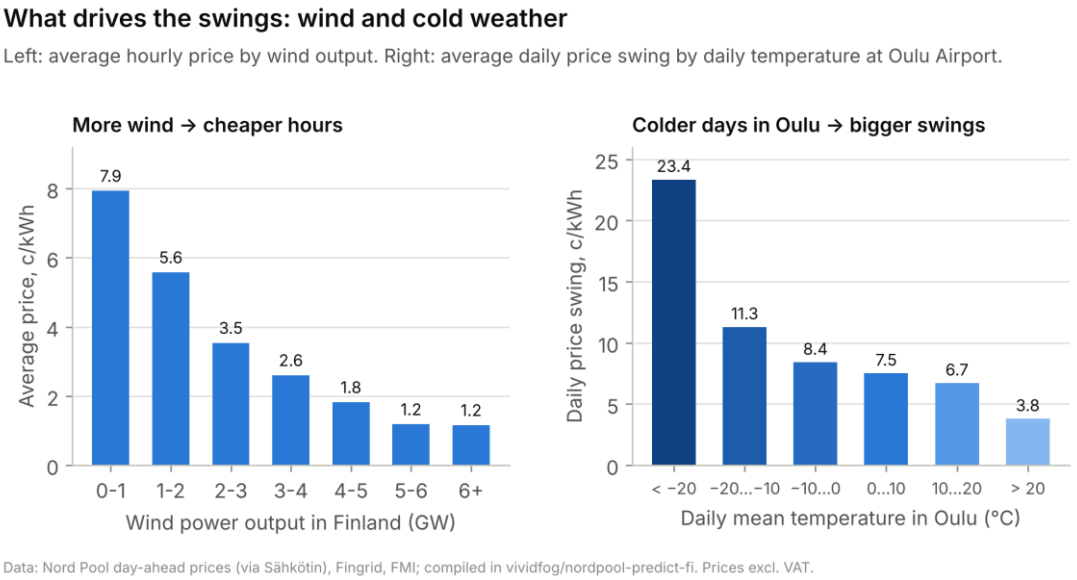


Figure 3. *Drivers of price swings: wind output and Oulu temperature.*

4.4 Value for households and market size

I modeled an Oulu family charging an EV with 7.5 kWh/day (about 13,800 km/year, Statistics Finland, 2025, 2026) and a 300-litre water heater with 8 kWh/day. The cheapest three and four hours save around €180 per year compared with charging on arrival (17–20) and a fixed 22–07 heating timer, steady in every period (Figure 4). A 10-kWh home battery can cost thousands of euros but only make roughly €250-300 per year. A service without hardware is a better entry point. Today, there are around 56,000 EV owners on spot contracts (33% of 169,458). The EV charging alone is therefore an indicative value pool of €7 million a year, or €22 million if all owners were on spot pricing.

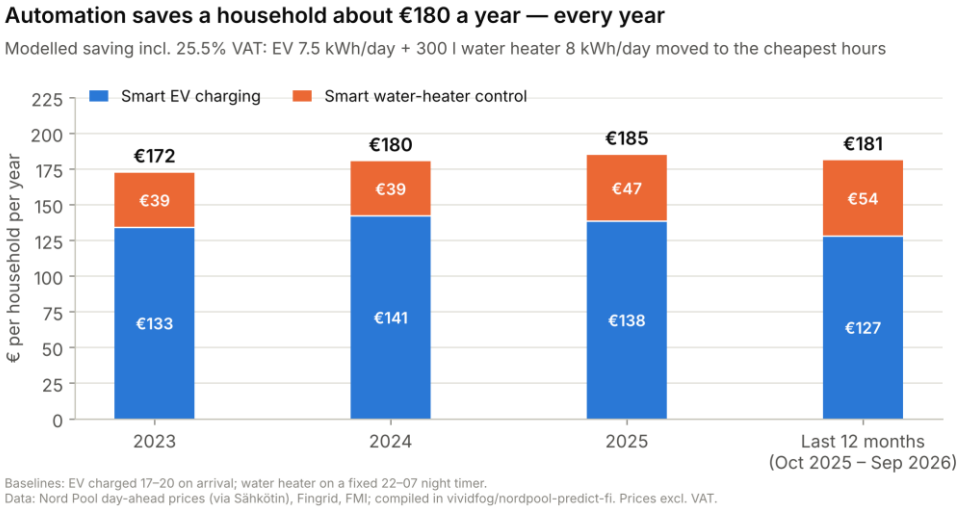


Figure 4. *Modelled annual saving per household.*

5. Feasibility Assessment and Recommendation

The data supports the opportunity on four tests: there is a value pool (huge daily swings), it is persistent (2023-2026), it is an opportunity that can be captured (predictable and disclosed in advance) and it is meaningful for customers (around €15 per month) in a fast-growing EV industry.

Risks do temper this result. Savings are minimal therefore the service needs to be inexpensive and easy enough to beat utility applications and do-it-yourself tools. Negative pricing hours reduced from 726 (2024) to 52 (January–September 2026), showing batteries and other flexible loads are absorbing excess wind. One-third of contracts only are linked to spot pricing. The savings are based on stated assumptions The data is from a hobby project (spot checks aligned with recorded events). 15-minute rates from October 2025 onward are averaged to hours.

Recommendation: This is a good opportunity to take to the next stage as a software first service. The next investment should be a small study on willingness to pay of Oulu homes with EVs and electric heating.

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